Happy Health – Patient 360 for the ACO

Data design for population health · draft for review

Ruben Sanchez – 11/07/2026

Patient 360 – who it serves and what it answers

One risk-ranked view of each patient across every facility they've touched, limited to each care manager's panel

Who uses it

- Care managers: patient detail, but only for the patients on their panel
- Care management leads: their team's panels, workload and outreach results
- ACO medical director and analysts: totals and trends without names, patient detail only with an approved, time-limited grant
- Primary care doctors (later): their attributed patients, inside Epic rather than a separate report

Questions it answers

- Who on my panel is highest risk, and who is rising?
- Who was admitted or seen in an emergency department in the last 30 days, at any Happy Health hospital or outside Happy Health?
- Which preventive and chronic-care gaps are open and due this year?
- Is the patient filling their medications?
- When did we last reach them, and when is their next appointment?

Explanation

The ACO wants one risk-stratified view of a patient across every facility they've touched, to target care-management outreach. That makes this the second consumer to depend on patient matching: a patient seen at an Epic hospital and at a MEDITECH hospital must appear once, with both histories.

The 360 is built after day 120 on the same foundations as the quality measures: the same silver data, the same patient links and the same shared dimensions. What's new is Medicare claims, which show care received outside Happy Health, and the care-management data: who is attributed to the ACO, which care manager owns each patient, and what outreach happened.

It is an operational view for outreach, not a filed number, so it always shows the latest identity and is rebuilt every night. That is the main design difference from the quality model.

Patient 360 – data model

Person-level facts for history, one wide summary row per patient for the worklist

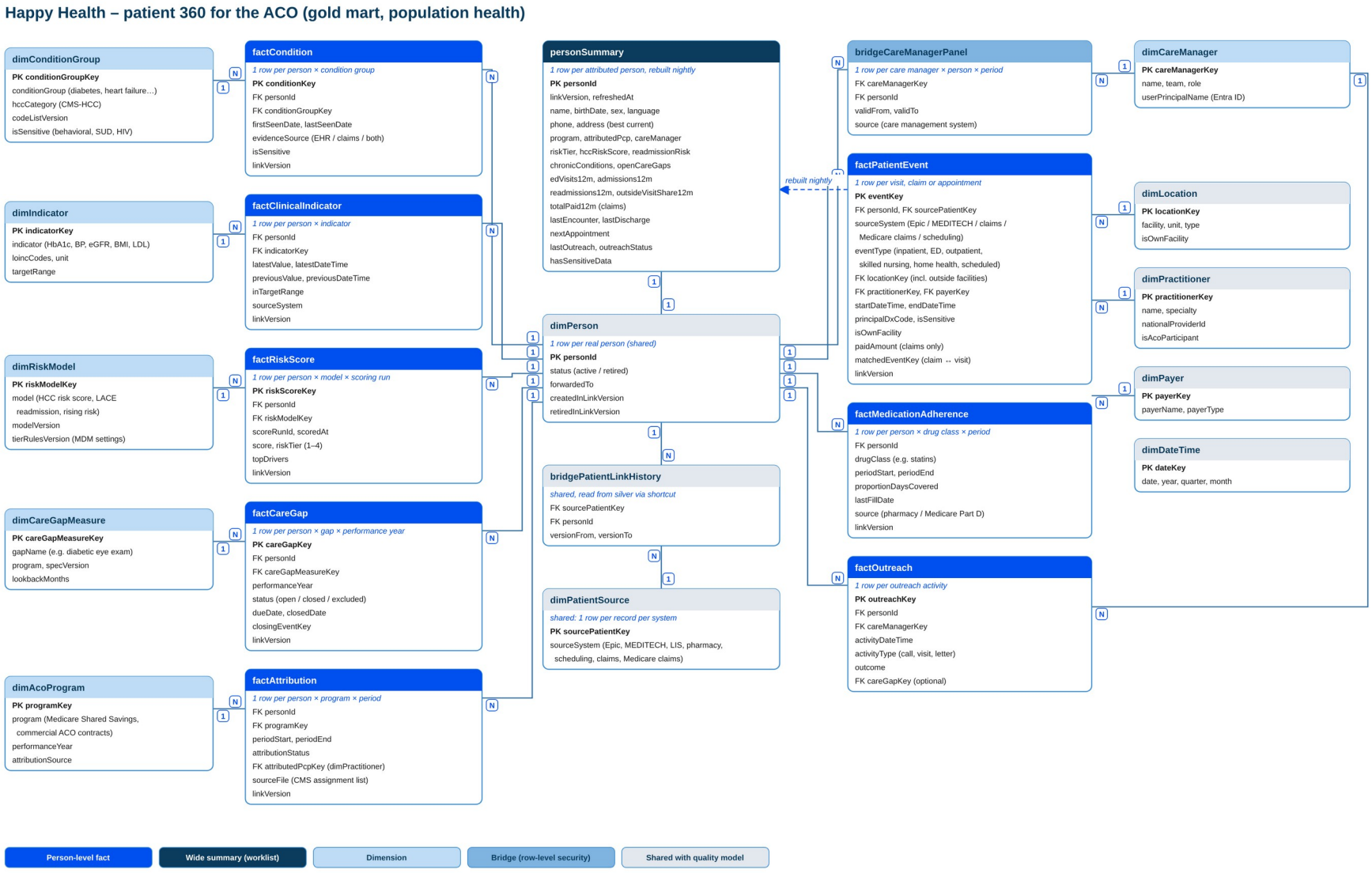


Figure 1. Patient 360 for the ACO – gold data model (population health)

How the model is shaped

- Centre: dimPerson, the one enterprise person, shared with the quality model
- Left: person-level facts for conditions, clinical indicators, risk scores, care gaps and ACO attribution, each with a small dimension
- Right: every visit, claim and appointment in factPatientEvent, plus medication adherence and outreach
- Top: personSummary, one wide row per attributed patient, rebuilt nightly for the care manager's worklist
- Row-level security comes from bridgeCareManagerPanel, which links each care manager to their patients

Explanation

The model has two shapes on purpose. The person-level facts keep history, so a care manager can drill into every visit, result and outreach call. personSummary is one wide row per attributed patient with the numbers a worklist needs: risk tier, open gaps, recent admissions and emergency visits, last outreach and next appointment. Reading one row per patient keeps the worklist fast, and because it is rebuilt from the facts every night it can't drift from them.

Grains: one row per person and condition group, per person and clinical indicator (latest and previous value), per person, risk model and scoring run, per person, care gap and performance year, and per person, program and attribution period. factPatientEvent has one row per visit, claim or appointment from any source.

Unlike the quality facts, these tables store the person ID directly, together with the link version used to build them. That is safe here because nothing is filed or frozen: when matching publishes a new link version, the 360 tables are rebuilt in full from silver rather than edited. The quality model keeps the source record on its facts because submitted numbers must never change.

Shared tables, shown in grey, come from the quality model and silver: dimPerson, dimLocation, dimPractitioner, dimPayer and dimDateTime, plus the patient link history and source records, read through a OneLake shortcut so there is one source of truth for identity. dimLocation gains outside facilities from Medicare claims, flagged with isOwnFacility. Date columns join to dimDateTime; those lines are left off the diagram to keep it readable.

A claim and the Happy Health visit it bills for are the same event. matchedEventKey links them (same person, facility and dates), so admissions aren't counted twice; claims with no matching visit are care received outside Happy Health.

Example: one patient in the worklist

John Jones (P-0005) from the synthetic workbook, as of 3 November 2026. Visits come from the workbook; conditions, scores and gaps are illustrative.

personSummary field	Value
personId · linkVersion	P-0005 · link version 8 (latest)
Source records	EPIC-10005 and MEDI-88005, joined in link version 7
Program · payer	Medicare Shared Savings (assumed) · Medicare (traditional)
riskTier · hccRiskScore · readmissionRisk	Tier 1 (highest) · 2.31 · LACE 14, high
chronicConditions	Heart failure, type 2 diabetes (illustrative)
admissions12m · readmissions12m	2 · 1 (Happy North 6–10 Jul, then Riverside 28 Jul–1 Aug)
edVisits12m	5 (Riverside 28 Jul; Happy Central 27 Aug; Happy West 8 and 12 Sep; Happy Valley 19 Sep)
openCareGaps	2: HbA1c test, diabetic eye exam (illustrative)
lastDischarge · lastEncounter	1 Aug 2026 Riverside · 19 Sep 2026 Happy Valley emergency department
lastOutreach · nextAppointment	2 Aug 2026 transition call, reached · 10 Nov 2026 primary care (illustrative)

Without the match in link version 7, the Riverside admission and emergency visit would belong to a different person, and this patient would look like a single July admission. The 360 shows four emergency visits in four weeks after a readmission, which is exactly the pattern outreach should act on.

Patient 360 – sources, lineage and refresh

Everything comes from silver; only Medicare claims, attribution and care-management data are new sources

Gold table	Built from (silver)	Original sources	Refresh
factPatientEvent	Encounters, claims, appointments	Epic Encounter; MEDITECH ADT visits; Happy Health claims system; Medicare claims files (Part A and B); scheduling	Nightly; Medicare claims monthly
factCondition	Diagnoses and problem lists, mapped to condition groups	Epic Condition; MEDITECH coded diagnoses (assumed addition); claim diagnoses (ICD-10-CM)	Nightly
factClinicalIndicator	Lab and vital-sign results (LOINC)	Epic Observation; MEDITECH results (ORU); lab system results	Nightly
factMedicationAdherence	Dispenses by drug class	Pharmacy dispenses (NDC); Medicare Part D claims	Nightly; Part D monthly
factRiskScore	Conditions, events, demographics	Calculated: HCC risk score, LACE readmission index, rising-risk rules	Weekly scoring run
factCareGap	Events, conditions, indicators, code lists	Calculated from dimCareGapMeasure specifications	Nightly
factAttribution	Attribution lists	CMS assignment list (quarterly); commercial payer rosters	Quarterly or monthly
factOutreach	Care-management activity	Care-management system (to confirm)	Nightly
bridgeCareManagerPanel	Panel assignments	Care-management system or ACO roster (to confirm)	Nightly
dimCareManager	Staff directory	Entra ID and HR	Nightly
personSummary	All 360 facts and dimPerson	Derived	Nightly, full rebuild

Explanation

Medicare claims are what make the view complete. For patients attributed to a Medicare ACO, CMS sends monthly claim files covering care anywhere, including outside hospitals, skilled nursing and home health, plus a quarterly list of which patients are assigned to the ACO. They land in bronze like any other source, are flattened in silver, and are matched to people by patient matching.

Claims arrive monthly and cover claims paid in the previous month, so outside care shows up weeks to months late. personSummary shows a 'data as of' date per source so care managers know how current each part is. Real-time admission alerts from the regional health information exchange are the later fix for that delay.

Risk scores and care gaps are calculated, not loaded. Like the quality measures, each scoring run records its model version and settings version, so a care manager can see why a patient was ranked where they were.

Patient 360 – identity, risk scores and care gaps

Always the latest identity; simple, explainable scores with their reasons shown

Identity

- The 360 always uses the latest published link version, and every table records which one it used
- A new link version triggers a full rebuild that night; tables are never patched
- Medicare claims carry the Medicare Beneficiary Identifier with name, birth date, sex and address; they go through Splink like any source, and the identifier is kept as a strong field for future matches
- The the Happy Health claims system's subscriber ID is never used alone, because it can belong to a family member
- After a split, the next rebuild moves events to the right person, and any outreach based on the wrong merge is reviewed with the privacy officer

Risk scores and tiers

- HCC risk score: CMS's model of expected cost from diagnoses and demographics, using the version CMS applies for the performance year
- Readmission risk: the LACE index (length of stay, acuity, other conditions, emergency visits) for recent discharges
- Rising risk: rules such as two or more emergency visits in 90 days, a new chronic condition, or medications filled less than 80% of the time
- Tier 1 to 4 from rules held in the master data management settings, versioned and approved like matching thresholds
- topDrivers stores the reasons, so the care manager sees why, not just a number

Explanation

We start with transparent scores that care managers can check and explain to a patient. A machine-learning model can come later, once outreach outcomes give it something to learn from, and it would slot in as another dimRiskModel row.

Care gaps reuse the measure pattern from the quality model: each gap has a versioned specification in dimCareGapMeasure, and a patient is in the gap's population when they qualify. The gap is closed when the qualifying event happens, such as an HbA1c result, and closingEventKey shows which event closed it. Gaps the patient qualifies for but hasn't closed are the outreach list.

Patient 360 – access and privacy

Panel-scoped detail for care managers; sensitive data and Medicare claims handled under their own rules

Who sees what

- Care managers: names, contact details and history, filtered by row-level security to rows in bridgeCareManagerPanel for their Entra ID sign-in and an open validity period
- Leads: a team role that sees their team's panels
- ACO analysts and medical director: a separate summary model without names, contact details or Medicare identifiers; patient detail needs an approved, time-limited grant
- Engineers: emergency access only, logged

Special rules

- Sensitive conditions (behavioral health, substance use treatment, HIV, reproductive health) are flagged with isSensitive and hidden by column-level security unless the role is authorized
- personSummary.hasSensitiveData only says restricted information exists, without showing it
- Medicare claims are used only for ACO-assigned patients and care coordination, under the ACO's data use agreement with CMS
- CMS already leaves out patients who declined data sharing and substance-use treatment claims; we never try to rebuild what was withheld

Explanation

This applies the same governance as the rest of the platform: Purview labels drive masking, Entra ID groups and OneLake security enforce roles, every query is logged in the audit layer, and access is reviewed quarterly.

The panel table is the key control. If a care manager leaves or a patient moves to another team, the old row is closed with validTo and access ends at the next refresh. A false merge is the biggest privacy risk here, because it could show a care manager another person's history; that is why matching thresholds start cautious and wrong merges trigger a privacy review.

Before care managers use it for self-service, the 360 semantic model goes through the Power BI Center of Excellence certification: definitions, row-level security, sensitivity labels, speed and ownership.

Patient 360 – delivery after day 120, open questions and risks

Three releases, each useful on its own

Release	What ships	Value to the ACO
1 (days 121–165)	Attribution, panels and row-level security; Happy Health visits from Epic and MEDITECH; personSummary worklist with admissions, emergency visits and next appointment	Care managers see who on their panel was admitted or seen in an emergency department at any Happy Health hospital
2 (days 166–210)	Medicare claims, outside facilities and cost; conditions; clinical indicators; care gaps	The full picture across every facility, and the outreach list of open gaps
3 (days 211–270)	Risk scores and tiers; medication adherence; outreach tracking; option for real-time admission alerts	Ranked worklists, and proof of which outreach worked

Open questions

- Which ACO programs: Medicare Shared Savings, another Medicare model, commercial contracts? How many attributed patients?
- Does Happy Health already receive the Medicare claim files, where are they stored, and what does the data use agreement allow?
- Where do panel assignments and outreach notes live today: Epic's population health module, an ACO vendor platform or spreadsheets?
- Does the ACO use a vendor risk score today that we should keep, compare against, or replace?
- Which care gaps matter most this performance year?
- Which conditions count as sensitive, and which roles may see them?
- Is a regional health information exchange available for real-time admission alerts?

Risks

- Claims arrive weeks to months late: show 'data as of' dates and add exchange alerts in release 3
- Attribution changes every quarter: keep attribution history so worklists and results explain who joined and left
- A false merge exposes another person's history: cautious thresholds, steward review, privacy review on every split
- Too many patients in the top tier makes the list useless: tune tier rules with care managers, track tier sizes